Text as Data

Unsupervised learning

Petro Tolochko

CEU

Fall 2026

Today

- What unsupervised models can help us discover
- Clustering and the choices behind similarity
- Topic models: LDA and the structural topic model
- Validation and text scaling

Supervised and unsupervised learning

- Supervised learning uses labelled examples to predict an outcome.
- Unsupervised learning describes patterns in data without training labels.
- The same texts may support different groupings or dimensions, depending on the representation and model.

One target: a parameter vector

In a model-based account, a vector $\boldsymbol{\theta}$ describes a particular feature of the world that we want to learn about.

$$\text{world and data-generating process} \quad \longrightarrow \quad \boldsymbol{\theta} \quad \longrightarrow \quad \widehat{\boldsymbol{\theta}}$$

- $\boldsymbol{\theta}$ is the target; $\widehat{\boldsymbol{\theta}}$ is our estimate from data.
- The target is defined by the question and model. Calling it a single target does not mean the whole of reality has one description.

Approximating a target

If the target $\boldsymbol{\theta}$ is well defined, we can ask how close an estimate comes:

$$L(\boldsymbol{\theta}, \widehat{\boldsymbol{\theta}}) = \|\boldsymbol{\theta} - \widehat{\boldsymbol{\theta}}\|^2.$$

- For example, parameters might describe how features relate to a specified outcome.
- In supervised learning, observed labels also let us assess predictions on held-out examples.
- Predictive accuracy and recovery of the true parameters are different goals; one does not guarantee the other.

What changes without labels?

- With a specified label, we can compare predicted and observed outcomes.
- In an exploratory task, there may be no predefined label or agreed target structure to recover.
- A model instead finds a structure $\widehat{S}$ in the observed texts X according to its objective.

$$X \quad \longrightarrow \quad \widehat{S}$$

The same texts, different structures

Consider a collection of parliamentary speeches. We could organize them by:

- **Policy area:** economy, defence, education.
- **Political position:** government versus opposition, or a left-right dimension.
- **Style:** procedural language versus emotional appeals.

These structures can coexist. Each answers a different question and may require different features and methods.

A single correct organization?

- Assuming one true θ for the entire collection can suggest there is only one correct way to organize it.
- Unsupervised methods can uncover several meaningful structures rather than approximating one universally correct classification.
- A specific unsupervised model can still assume a particular latent structure and estimate its parameters.

What makes a discovered structure meaningful?

- Does it answer a useful substantive question?
- Do representative texts support the proposed interpretation?
- Does it remain recognizable across reasonable model and preprocessing choices?
- Does relevant external evidence support the interpretation?

Different structures can be useful, but that does not make every grouping equally convincing.

Measurement and model objectives

An objective function describes what a model optimizes. For example, k-means seeks compact clusters.

- A research question gives those patterns a possible interpretation.
- A good optimization score alone cannot establish that a cluster or topic measures a social concept.
- Computing performance and substantive interpretation matter in both computer science and social science, though their emphasis can differ.

Discovery and interpretation

- Models can reveal unexpected patterns as well as examine theoretically motivated questions.
- A topic or cluster is a result to investigate, not automatically a real-world category.
- Examine representative documents and check stability, alternative explanations, and relevant external evidence.

What is Measurement?

- Measurement = the process of assigning numbers (or symbols) to phenomena according to rules.
- Involves three linked steps:
  - **Conceptualization:** What do we want to capture?
  - **Operationalization:** How can we represent it empirically?
  - **Quantification:** How do we express it numerically?
- Measurement always implies a mapping between *theoretical constructs* and *observed data.*

What Is an Objective Function?

- In machine learning, the **objective function** defines what the model tries to achieve.
- It translates a goal into something computable.
- Examples:
  - Linear regression: minimize squared error $(y - \hat{y})^2$
  - K-means: minimize within-cluster variance
  - PCA: minimize reconstruction error
- **Think of it as:** the model's "definition of success."

Why the Objective Function Matters

- Defines what the model "cares about".
- Different objectives $\Rightarrow$ different structures discovered.
- If the objective does not align with theory, results can mislead.

Good measurement = alignment between objective and theory.

Prediction vs Inference

- **Prediction:** How well does the model reproduce unseen data?
- **Inference:** What do parameters tell us about the world?
- Large parameter spaces $\Rightarrow$ poor inference, good prediction

Problems

- **Translation of social science concepts** Machine learning measures patterns, not constructs.
- **Connecting methods to theory** Models optimize statistical objectives, not theoretical meanings.
- **Difficult to understand what is being measured** Without a clear mapping between theory and model, $\hat{S}$ may have no interpretable referent.

Families of Unsupervised Methods

- **Clustering:**
  Find groups of similar items.
  Examples: K-Means, Hierarchical Clustering.

- **Topic Modeling:**
  Find latent themes in text.
  Examples: Latent Dirichlet Allocation (LDA), Structural Topic Model (STM).

- **Scaling:**
  Place items along latent dimensions.
  Example: Wordfish (Slapin & Proksch, 2008).

K-Means Clustering

- Partitions data into K non-overlapping clusters
- Simple(ish) algorithmic method

K-Means Clustering

$$C_1, C_2, \ldots, C_K$$

K-Means Clustering

$$C_1, C_2, \ldots, C_K$$

$$C_1 \cup C_2 \cup \ldots \cup C_K = \{1, \ldots, n\}$$

K-Means Clustering

$$C_1, C_2, \ldots, C_K$$

$$C_1 \cup C_2 \cup \ldots \cup C_K = \{1, \ldots, n\}$$

$$C_k \cap C_{k'} = \varnothing \quad \text{for all } k \neq k'$$

K-Means Clustering

$$C_1, C_2, \ldots, C_K$$

$$C_1 \cup C_2 \cup \ldots \cup C_K = \{1, \ldots, n\}$$

$$C_k \cap C_{k'} = \varnothing \quad \text{for all } k \neq k'$$

Each observation belongs to exactly one cluster.
Clusters are non-overlapping and collectively exhaustive.

Objective Function

Goal: Find K clusters that minimize within-cluster variation.

Objective Function

Goal: Find K clusters that minimize within-cluster variation.

$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

Objective Function

Goal: Find K clusters that minimize within-cluster variation.

$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

where μ_k is the centroid (mean) of cluster C_k:

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

Objective Function

Goal: Find K clusters that minimize within-cluster variation.

$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

where μ_k is the centroid (mean) of cluster C_k:

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

Intuition: assign points to clusters so that each cluster is as compact as possible.

K-Means Algorithm

Algorithm: Iterative refinement of cluster assignments.

K-Means Algorithm

Algorithm: Iterative refinement of cluster assignments.

1. Choose the number of clusters K.

K-Means Algorithm

Algorithm: Iterative refinement of cluster assignments.

1. Choose the number of clusters K.
2. Randomly assign each observation to one of the K clusters.

K-Means Algorithm

Algorithm: Iterative refinement of cluster assignments.

1. Choose the number of clusters K.
2. Randomly assign each observation to one of the K clusters.
3. Compute the centroid μ_k for each cluster:

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

K-Means Algorithm

Algorithm: Iterative refinement of cluster assignments.

1. Choose the number of clusters K.
2. Randomly assign each observation to one of the K clusters.
3. Compute the centroid μ_k for each cluster:

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

4. Reassign each observation to the cluster with the nearest centroid.

K-Means Algorithm

Algorithm: Iterative refinement of cluster assignments.

1. Choose the number of clusters K.
2. Randomly assign each observation to one of the K clusters.
3. Compute the centroid μ_k for each cluster:

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

4. Reassign each observation to the cluster with the nearest centroid.
5. Repeat steps 3–4 until assignments do not change.

K-Means Algorithm

Algorithm: Iterative refinement of cluster assignments.

1. Choose the number of clusters K.
2. Randomly assign each observation to one of the K clusters.
3. Compute the centroid μ_k for each cluster:

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

4. Reassign each observation to the cluster with the nearest centroid.
5. Repeat steps 3–4 until assignments do not change.

K-Means converges to a local optimum, not necessarily the global one.

Choosing the Number of Clusters

- **Elbow method:** plot WCSS vs. K
- **Silhouette score:** average distance to own cluster vs others
- **Theory-informed:** choose K based on expected structure

Cluster Algorithms Validation

- **Data assumptions:** check if clustering assumptions match data generation.

Cluster Algorithms Validation

- **Data assumptions:** check if clustering assumptions match data generation.
- **Internal validity:** best results for the data (e.g., cohesion, separation).

Cluster Algorithms Validation

- **Data assumptions:** check if clustering assumptions match data generation.
- **Internal validity:** best results for the data (e.g., cohesion, separation).
- **External validity:** alignment with pre-existing understanding or labels.

Cluster Algorithms Validation

- **Data assumptions:** check if clustering assumptions match data generation.
- **Internal validity:** best results for the data (e.g., cohesion, separation).
- **External validity:** alignment with pre-existing understanding or labels.
- **Cross-validity:** stability across similar datasets or time periods.

Cluster Algorithms Validation

- **Data assumptions:** check if clustering assumptions match data generation.
- **Internal validity:** best results for the data (e.g., cohesion, separation).
- **External validity:** alignment with pre-existing understanding or labels.
- **Cross-validity:** stability across similar datasets or time periods.
- **You are the validation method.**
  Interpretation and theory provide the ultimate validation.
- Validation moves from *data–fit* to *meaning–fit.*

Topic Models

Topic Modeling

- **Goal:** discover latent themes or "topics" in a collection of documents.

Topic Modeling

- **Goal:** discover latent themes or "topics" in a collection of documents.
- Topic modeling is a **family of models**, not a single algorithm.

Topic Modeling

- **Goal:** discover latent themes or "topics" in a collection of documents.
- Topic modeling is a **family of models**, not a single algorithm.
- **Latent Dirichlet Allocation (LDA)** is one of several topic models.

Topic Modeling

- **Goal:** discover latent themes or "topics" in a collection of documents.
- Topic modeling is a **family of models**, not a single algorithm.
- **Latent Dirichlet Allocation (LDA)** is one of several topic models.
- Other approaches include:
  - Latent Semantic Analysis (LSA)
  - Singular Value Decomposition (SVD)
  - Clustering-based topic discovery

Latent Dirichlet Allocation (LDA)

- A **Bayesian generative hierarchical model** for discovering topics in text.

Latent Dirichlet Allocation (LDA)

- A **Bayesian generative hierarchical model** for discovering topics in text.
- Originally introduced to model *population structure* in genetics (Pritchard, Stephens & Donnelly, 2000).

Latent Dirichlet Allocation (LDA)

- A **Bayesian generative hierarchical model** for discovering topics in text.
- Originally introduced to model *population structure* in genetics (Pritchard, Stephens & Donnelly, 2000).
- Adapted to text analysis as a probabilistic topic model (Blei, Ng & Jordan, 2003).

Latent Dirichlet Allocation (LDA)

- A **Bayesian generative hierarchical model** for discovering topics in text.
- Originally introduced to model *population structure* in genetics (Pritchard, Stephens & Donnelly, 2000).
- Adapted to text analysis as a probabilistic topic model (Blei, Ng & Jordan, 2003).
- Each document is modeled as a *mixture of topics*, and each topic as a *distribution over words*.

Latent Dirichlet Allocation (LDA)

- Goal: discover latent topics that generate observed words.
- Assumption: documents are mixtures of topics; topics are distributions of words.

Mixture Models

- **Idea:** observations come from a combination of several underlying distributions.

Mixture Models

- **Idea:** observations come from a combination of several underlying distributions.
- Each distribution represents a **latent group or component**.

Mixture Models

- **Idea:** observations come from a combination of several underlying distributions.
- Each distribution represents a **latent group or component**.
- The probability of each component is weighted by a **mixture proportion**.

Mixture Models

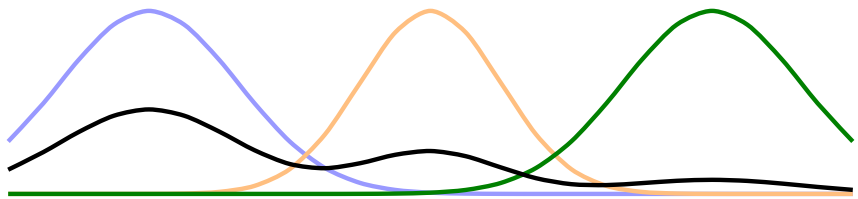
- **Idea:** observations come from a combination of several underlying distributions.
- Each distribution represents a **latent group or component**.
- The probability of each component is weighted by a **mixture proportion**.
- Unlike K-means, mixture models use *soft assignment:*
  each observation belongs to components with certain probabilities.

Mixture Models

- **Idea:** observations come from a combination of several underlying distributions.
- Each distribution represents a **latent group or component**.
- The probability of each component is weighted by a **mixture proportion**.
- Unlike K-means, mixture models use *soft assignment:* each observation belongs to components with certain probabilities.

Example: a document can be 70% about politics, 20% about health, 10% about sports.

Mixture Model



Observed data = weighted combination of latent distributions

Hierarchical Models

$$y = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k + \varepsilon$$

Hierarchical Models

$$y = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k + \varepsilon$$

In a standard regression model, coefficients β are fixed unknowns.
We estimate them directly from the data.

Hierarchical Models

$$y = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k + \varepsilon$$

In a standard regression model, coefficients β are fixed unknowns.
We estimate them directly from the data.

In hierarchical (Bayesian) models, coefficients are random variables with their own probability distributions.

Hierarchical Models

$$
\begin{aligned}
y &\sim Normal(\mu, \sigma) \\
\mu &= \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k \\
\beta &\sim Normal(0, 5) \\
\sigma &\sim Exponential(1)
\end{aligned}
$$

Hierarchical Models

$$
\begin{aligned}
y &\sim Normal(\mu, \sigma) \\
\mu &= \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k \\
\beta &\sim Normal(0, 5) \\
\sigma &\sim Exponential(1)
\end{aligned}
$$

Likelihood: $y \sim Normal(\mu, \sigma)$: how data are generated.
Priors: β, σ have their own distributions.
The model is hierarchical because parameters depend on other parameters.

Hierarchical Models: Hyperparameters

$$\beta \sim Normal(0, 5)$$
$$\sigma \sim Exponential(1)$$

Hierarchical Models: Hyperparameters

$$\beta \sim Normal(0, 5)$$
$$\sigma \sim Exponential(1)$$

The numbers $(0, 5)$ and (1) are **hyperparameters**:
they define the prior distributions for model parameters.

Hierarchical Models: Hyperparameters

$$\beta \sim Normal(0, 5)$$
$$\sigma \sim Exponential(1)$$

The numbers $(0, 5)$ and (1) are **hyperparameters**:
they define the prior distributions for model parameters.

In LDA, the Dirichlet priors α and β play this same role.
They control how concentrated or diffuse the topic and document distributions are.

LDA Formalized

Generative process:

1. For each topic $k \in \{1, \ldots, K\}$:

$$\theta_k \sim \text{Dirichlet}(\alpha)$$

(distribution of words in topic k)

LDA Formalized

Generative process:

1. For each topic $k \in \{1, \ldots, K\}$:

$$\theta_k \sim \text{Dirichlet}(\alpha)$$

(distribution of words in topic k)

2. For each document $d \in \{1, \ldots, D\}$:

$$\eta_d \sim \text{Dirichlet}(\beta)$$

(distribution of topics in document d)

LDA Formalized

Generative process:

1. For each topic $k \in \{1, \ldots, K\}$:

$$\theta_k \sim \text{Dirichlet}(\alpha)$$

(distribution of words in topic k)

2. For each document $d \in \{1, \ldots, D\}$:

$$\eta_d \sim \text{Dirichlet}(\beta)$$

(distribution of topics in document d)

3. For each word w in document d:

$$z_{d,w} \sim \text{Multinomial}(\eta_d) \quad \text{(topic assignment)}$$
$$x_{d,w} \sim \text{Multinomial}(\theta_{z_{d,w}}) \quad \text{(observed word)}$$

LDA Formalized

Generative process:

1. For each topic $k \in \{1, \ldots, K\}$:

$$\theta_k \sim \text{Dirichlet}(\alpha)$$

(distribution of words in topic k)

2. For each document $d \in \{1, \ldots, D\}$:

$$\eta_d \sim \text{Dirichlet}(\beta)$$

(distribution of topics in document d)

3. For each word w in document d:

$$z_{d,w} \sim \text{Multinomial}(\eta_d) \quad \text{(topic assignment)}$$
$$x_{d,w} \sim \text{Multinomial}(\theta_{z_{d,w}}) \quad \text{(observed word)}$$

Goal: infer the latent variables θ, η, and z given observed words x.

LDA Formalized

Generative process:

1. For each topic $k \in \{1, \ldots, K\}$:

$$\theta_k \sim \text{Dirichlet}(\alpha)$$

(distribution of words in topic k)

2. For each document $d \in \{1, \ldots, D\}$:

$$\eta_d \sim \text{Dirichlet}(\beta)$$

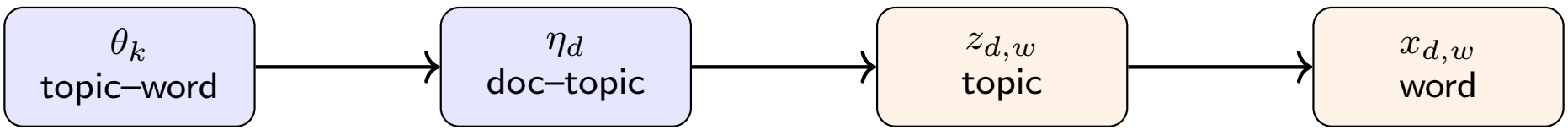
(distribution of topics in document d)

3. For each word w in document d:

$$z_{d,w} \sim \text{Multinomial}(\eta_d) \quad \text{(topic assignment)}$$
$$x_{d,w} \sim \text{Multinomial}(\theta_{z_{d,w}}) \quad \text{(observed word)}$$

Goal: infer the latent variables θ, η, and z given observed words x.
Formally: $P(\theta, \eta, z \mid x, \alpha, \beta)$

LDA Generative Process



Model first draws topic–word distributions, then document–topic mixtures, assigns topics to words, and generates observed words.

LDA in Words

1. For each topic k, draw word distribution $\theta_k \sim Dir(\alpha)$
2. For each document d, draw topic mixture $\eta_d \sim Dir(\beta)$
3. For each word:
   - Choose topic $z_{d,w} \sim Mult(\eta_d)$
   - Choose word $x_{d,w} \sim Mult(\theta_{z_{d,w}})$

Dirichlet Distribution

- A **distribution over distributions.**

Dirichlet Distribution

- A **distribution over distributions.**
- It describes random vectors that sum to 1:

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_K), \quad \text{where } \sum_{k=1}^{K} \theta_k = 1.$$

Dirichlet Distribution

- A **distribution over distributions.**
- It describes random vectors that sum to 1:

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_K), \quad \text{where } \sum_{k=1}^{K} \theta_k = 1.$$

- Parameterized by $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$, which control how concentrated or spread out the probabilities are.

Dirichlet Distribution

- A **distribution over distributions.**
- It describes random vectors that sum to 1:

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_K), \quad \text{where } \sum_{k=1}^{K} \theta_k = 1.$$

- Parameterized by $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$, which control how concentrated or spread out the probabilities are.
- **High** α**:** all θ_k similar (uniform).
  Low α**:** few θ_k dominate (sparse distribution).

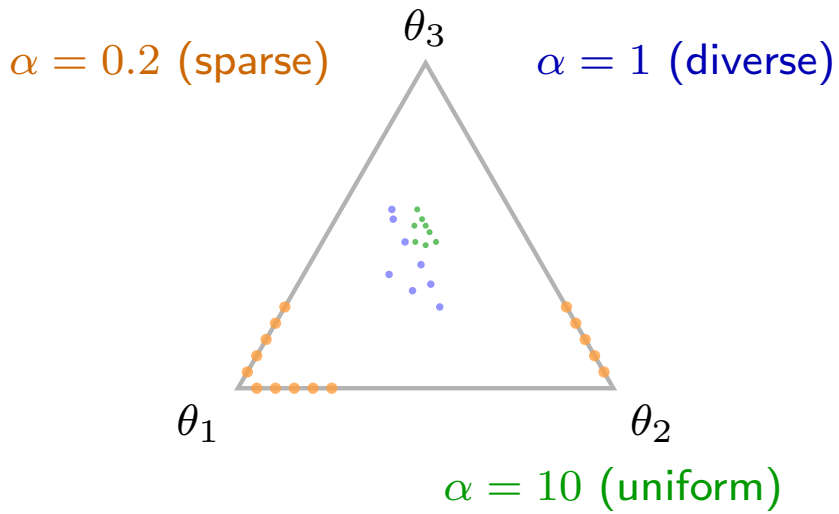
Dirichlet Distribution

- A **distribution over distributions.**
- It describes random vectors that sum to 1:

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_K), \quad \text{where } \sum_{k=1}^{K} \theta_k = 1.$$

- Parameterized by $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$, which control how concentrated or spread out the probabilities are.
- **High** α**:** all θ_k similar (uniform).
  Low α**:** few θ_k dominate (sparse distribution).
- In LDA, Dirichlet priors control how concentrated topics or words are.

Visualizing the Dirichlet Distribution



Smaller $\alpha \rightarrow$ more peaked; larger $\alpha \rightarrow$ more uniform.
Each dot is a draw $(\theta_1, \theta_2, \theta_3)$ lying on the simplex.

Dirichlet in LDA

Variable	**Drawn from**	**Meaning**
η_d	Dirichlet(β)	Topic distribution for document d
θ_k	Dirichlet(α)	Word distribution for topic k

Dirichlet priors define the diversity or sparsity of topics and words.

Dirichlet priors in this notation

In the notation used on the preceding slides:

- $\eta_d \sim \text{Dirichlet}(\beta)$: document d's distribution over topics.
- $\theta_k \sim \text{Dirichlet}(\alpha)$: topic k's distribution over words.

For symmetric priors with the number of categories fixed, smaller values tend to produce sparser draws and larger values more even draws.
Notation varies: many references use α for document-topic and β for topic-word priors. Check the definition in the software or paper.

From LDA to STM

- **LDA:** all documents share the same prior over topics.

From LDA to STM

- **LDA:** all documents share the same prior over topics.
- **STM (Structural Topic Model):** extends LDA by allowing topic prevalence and content to vary with **document covariates**.

From LDA to STM

- **LDA:** all documents share the same prior over topics.
- **STM (Structural Topic Model):** extends LDA by allowing topic prevalence and content to vary with **document covariates**.
- Covariates can affect:
  - **Topic prevalence** — how much each topic appears in a document.
  - **Topic content** — which words are used to discuss that topic.

From LDA to STM

- **LDA:** all documents share the same prior over topics.
- **STM (Structural Topic Model):** extends LDA by allowing topic prevalence and content to vary with **document covariates**.
- Covariates can affect:
  - **Topic prevalence** — how much each topic appears in a document.
  - **Topic content** — which words are used to discuss that topic.
- Uses a **Logistic-Normal** prior instead of a Dirichlet:

$$\eta_d \sim \text{LogisticNormal}(\mu_d, \Sigma), \quad \mu_d = X_d\gamma$$

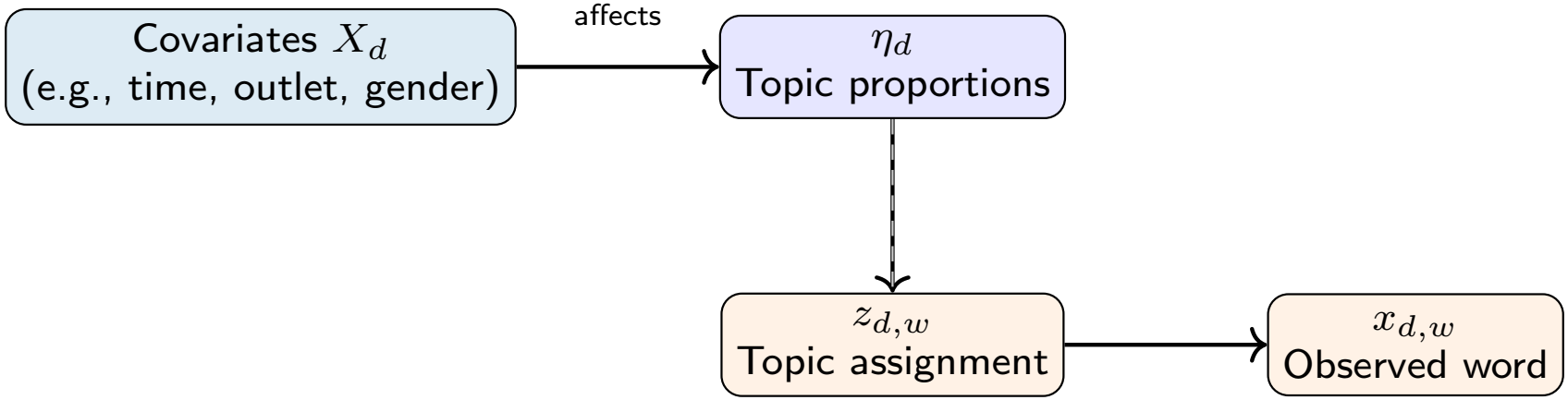
From LDA to STM

- **LDA:** all documents share the same prior over topics.
- **STM (Structural Topic Model):** extends LDA by allowing topic prevalence and content to vary with **document covariates**.
- Covariates can affect:
  - **Topic prevalence** — how much each topic appears in a document.
  - **Topic content** — which words are used to discuss that topic.
- Uses a **Logistic-Normal** prior instead of a Dirichlet:

$$\eta_d \sim \text{LogisticNormal}(\mu_d, \Sigma), \quad \mu_d = X_d\gamma$$

- Enables inference: how topics vary by time, source, gender, ideology, etc.

How STM Extends LDA



STM extends LDA by adding document-level structure:
*Covariates influence which topics appear (***prevalence***) and how topics are expressed (***content***).*

Why Validate Unsupervised Models?

- No ground truth $\Rightarrow$ model can always find patterns.

Why Validate Unsupervised Models?

- No ground truth $\Rightarrow$ model can always find patterns.
- Validation checks whether discovered structure reflects:
  - the *data* (internal coherence),
  - the *world* (external correspondence),
  - or the *theory* (interpretive meaning).

Why Validate Unsupervised Models?

- No ground truth $\Rightarrow$ model can always find patterns.
- Validation checks whether discovered structure reflects:
  - the *data* (internal coherence),
  - the *world* (external correspondence),
  - or the *theory* (interpretive meaning).
- Without validation, unsupervised results are **artifacts of the algorithm**.

Validation of Unsupervised Models

- **Data assumptions:** check that model assumptions match the data structure.
  Example: K-means assumes spherical clusters, LDA assumes word exchangeability.

Validation of Unsupervised Models

- **Data assumptions:** check that model assumptions match the data structure.
  Example: K-means assumes spherical clusters, LDA assumes word exchangeability.
- **Internal validity:** evaluate fit within the data (e.g., cohesion, silhouette, WCSS, topic coherence).

Validation of Unsupervised Models

- **Data assumptions:** check that model assumptions match the data structure.
  Example: K-means assumes spherical clusters, LDA assumes word exchangeability.
- **Internal validity:** evaluate fit within the data (e.g., cohesion, silhouette, WCSS, topic coherence).
- **External validity:** compare with known or manually coded classifications, human judgments, or metadata.

Validation of Unsupervised Models

- **Data assumptions:** check that model assumptions match the data structure.
  Example: K-means assumes spherical clusters, LDA assumes word exchangeability.
- **Internal validity:** evaluate fit within the data (e.g., cohesion, silhouette, WCSS, topic coherence).
- **External validity:** compare with known or manually coded classifications, human judgments, or metadata.
- **Cross-validity:** check stability across samples, time periods, or model runs.

Validation of Unsupervised Models

- **Data assumptions:** check that model assumptions match the data structure.
  Example: K-means assumes spherical clusters, LDA assumes word exchangeability.
- **Internal validity:** evaluate fit within the data (e.g., cohesion, silhouette, WCSS, topic coherence).
- **External validity:** compare with known or manually coded classifications, human judgments, or metadata.
- **Cross-validity:** check stability across samples, time periods, or model runs.
- **Interpretive validity:** the researcher's qualitative assessment of meaning — *You are the validation method.*

Validating Topic Models

- **Semantic validity:** do top words within each topic form a coherent theme?
  (often checked with human evaluation)

Validating Topic Models

- **Semantic validity:** do top words within each topic form a coherent theme?
  (often checked with human evaluation)
- **Convergent validity:** do topic proportions correlate with external variables
  or manual codings for the same documents?
  (e.g., Guo et al., 2016)

Validating Topic Models

- **Semantic validity:** do top words within each topic form a coherent theme?
  (often checked with human evaluation)
- **Convergent validity:** do topic proportions correlate with external variables
  or manual codings for the same documents?
  (e.g., Guo et al., 2016)
- **Quantitative metrics:**
  - *Topic coherence* (e.g., NPMI, UMass, UCI)
  - *Held-out likelihood* or *perplexity*

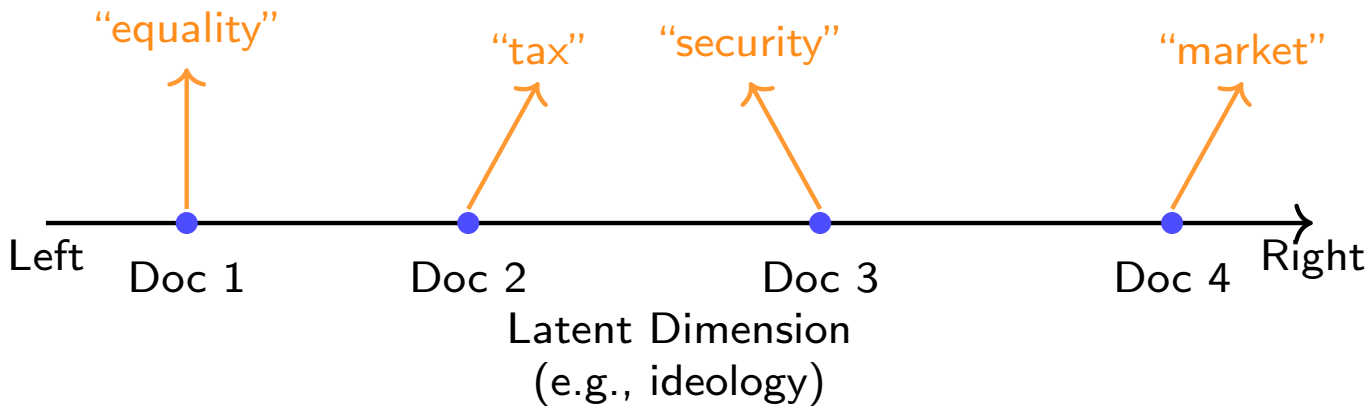
Validating Topic Models

- **Semantic validity:** do top words within each topic form a coherent theme?
  (often checked with human evaluation)
- **Convergent validity:** do topic proportions correlate with external variables
  or manual codings for the same documents?
  (e.g., Guo et al., 2016)
- **Quantitative metrics:**
  - *Topic coherence* (e.g., NPMI, UMass, UCI)
  - *Held-out likelihood* or *perplexity*
- **Human-in-the-loop validation:**
  Experts or crowdsourced coders rate interpretability and label consistency.

Text Scaling Methods

- Place documents on latent dimensions (e.g., ideology).
- Use discriminating word frequencies as anchors.

Text Scaling



Documents are positioned along a latent dimension according to word usage. Discriminating words push documents toward one end or the other.

Assumptions of Text Scaling Methods

- **Unidimensional structure:** texts vary along one latent continuum (e.g., ideology).

Assumptions of Text Scaling Methods

- **Unidimensional structure:** texts vary along one latent continuum (e.g., ideology).
- **Conditional independence:** words are independent given a document's latent position.

Assumptions of Text Scaling Methods

- **Unidimensional structure:** texts vary along one latent continuum (e.g., ideology).
- **Conditional independence:** words are independent given a document's latent position.
- **Stable word meanings:** each word has the same directional association across texts.

Assumptions of Text Scaling Methods

- **Unidimensional structure:** texts vary along one latent continuum (e.g., ideology).
- **Conditional independence:** words are independent given a document's latent position.
- **Stable word meanings:** each word has the same directional association across texts.
- **Comparability:** all documents are drawn from the same domain or context.

Assumptions of Text Scaling Methods

- **Unidimensional structure:** texts vary along one latent continuum (e.g., ideology).
- **Conditional independence:** words are independent given a document's latent position.
- **Stable word meanings:** each word has the same directional association across texts.
- **Comparability:** all documents are drawn from the same domain or context.
- **Signal vs. topic:** variation in word use reflects position, not topic differences.

Assumptions of Text Scaling Methods

- **Unidimensional structure:** texts vary along one latent continuum (e.g., ideology).
- **Conditional independence:** words are independent given a document's latent position.
- **Stable word meanings:** each word has the same directional association across texts.
- **Comparability:** all documents are drawn from the same domain or context.
- **Signal vs. topic:** variation in word use reflects position, not topic differences.
- **Scale invariance:** only relative positions matter, the scale's origin is arbitrary.

Wordfish as a Hierarchical Model (Slapin & Proksch, 2008)

Likelihood:

$$y_{ik} \sim \text{Poisson}(\lambda_{ik})$$

$$\log \lambda_{ik} = \alpha_i + \psi_k + \beta_k \, \omega_i$$

Wordfish as a Hierarchical Model (Slapin & Proksch, 2008)

Likelihood:

$$y_{ik} \sim \text{Poisson}(\lambda_{ik})$$

$$\log \lambda_{ik} = \alpha_i + \psi_k + \beta_k \, \omega_i$$

Parameters:

- α_i — document-specific intercept (captures length / verbosity)
- ψ_k — word-specific intercept (baseline frequency)
- β_k — word-specific discrimination (strength of association with dimension)
- ω_i — document position on the latent scale

Wordfish as a Hierarchical Model (Slapin & Proksch, 2008)

Likelihood:

$$y_{ik} \sim \text{Poisson}(\lambda_{ik})$$

$$\log \lambda_{ik} = \alpha_i + \psi_k + \beta_k \, \omega_i$$

Parameters:

- α_i — document-specific intercept (captures length / verbosity)
- ψ_k — word-specific intercept (baseline frequency)
- β_k — word-specific discrimination (strength of association with dimension)
- ω_i — document position on the latent scale

Hierarchical structure:

$$\alpha_i \sim \text{Normal}(0, \sigma_\alpha)$$

$$\psi_k \sim \text{Normal}(0, \sigma_\psi)$$

$$\beta_k \sim \text{Normal}(0, \sigma_\beta)$$

$$\omega_i \sim \text{Normal}(0, 1)$$

Wordfish as a Hierarchical Model (Slapin & Proksch, 2008)

Likelihood:

$$y_{ik} \sim \text{Poisson}(\lambda_{ik})$$

$$\log \lambda_{ik} = \alpha_i + \psi_k + \beta_k \, \omega_i$$

Parameters:

- α_i — document-specific intercept (captures length / verbosity)
- ψ_k — word-specific intercept (baseline frequency)
- β_k — word-specific discrimination (strength of association with dimension)
- ω_i — document position on the latent scale

Hierarchical structure:

$$\alpha_i \sim \text{Normal}(0, \sigma_\alpha)$$

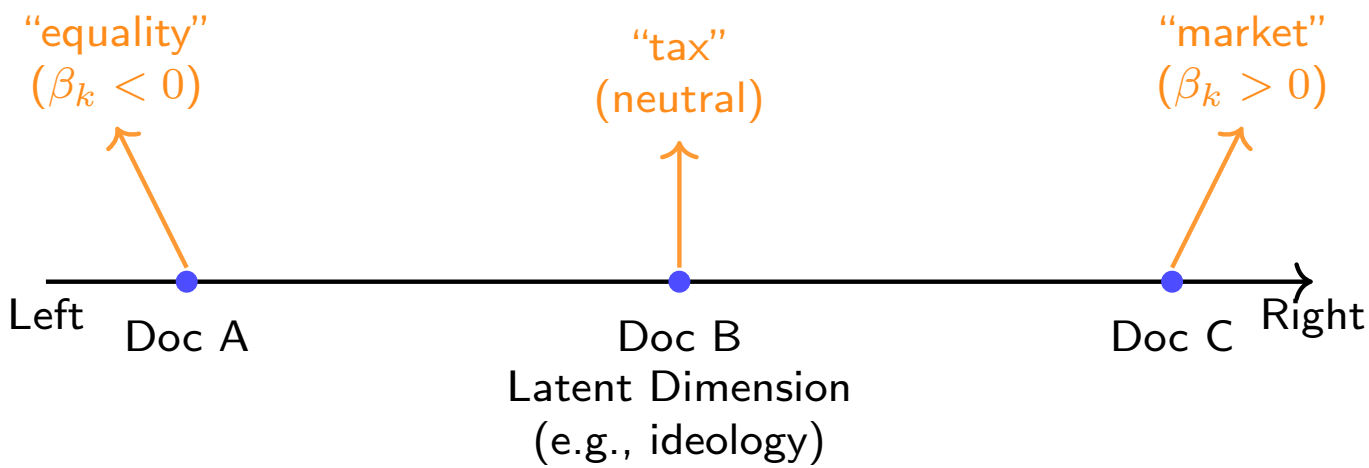
$$\psi_k \sim \text{Normal}(0, \sigma_\psi)$$

$$\beta_k \sim \text{Normal}(0, \sigma_\beta)$$

$$\omega_i \sim \text{Normal}(0, 1)$$

Wordfish estimates document positions ω_i and word discriminations β_k jointly.

How Wordfish Links Words and Documents



Words with positive β_k pull documents to the right; words with negative β_k pull them to the left.
Estimated ω_i summarize these directional tendencies.

Summary: Comparing Unsupervised Methods

	Goal	**Output**	**Example**
Clustering	Grouping	Cluster labels	K-means
Topic Modeling	Thematic discovery	Topic-word probs	LDA / STM
Scaling	Latent dimension	Position scores	Wordfish

What Unsupervised Learning Does (and Doesn't Do)

- **Discovers structure**, it does not define it.
- **Measures relationships**, not "truth".
- Depends on **assumptions about meaning, similarity, and latent space.**
- **Useful for exploration and theory building**, not just prediction.

Unsupervised learning helps us see structure we didn't know was there, but we still have to interpret what that structure means.

Each method is a different kind of measurement, and every measurement is a theoretical exercise.

Questions?